

Characters moving in time and space

Who was where, and when: tracking each character's movement through the places a narrative names, as an animated map.

The question

A narrative moves its people about. Asking where a character is at a given point in the story — and who else is there — turns a text into something closer to a historical record: a set of people, places and moments that can be mapped, compared, and checked against what is known from elsewhere.

This is the geocodable half of the question. Places that can be put on a map of the world are handled here; places that cannot — a kitchen, a forest, "the other side of the tracks" — belong to symbolic space, and have their own tool and their own TIPS.

What the tool does

1. Stanza's named-entity recogniser reads every sentence and picks out the PERSON entities and the place entities — GPE, LOC, CITY, COUNTRY, STATE_OR_PROVINCE, FAC.
2. Wherever a sentence names at least one person AND at least one place, the tool records the pair: this character, this place, this sentence, this document. A sentence with two people and two places yields four pairs.
3. The distinct place names are geocoded — turned into coordinates — through OpenStreetMap's Nominatim service.
4. Each character's places are put in narrative order and drawn as a path on a Leaflet map, with a timeline slider that plays the movement through.

The unit of "being somewhere" is therefore CO-OCCURRENCE IN A SENTENCE. The tool does not parse who went where; it records that a person and a place were named together. See "What it does not tell you" below — this is the caveat that matters most.

Where to find it

- **GIS GUI.** the checkbox "MAP characters moving in time and geocodable space". This is the tool's home.
- **Corpus Profiler.** runs it as one of its automated analyses and puts the map in the report.
- **Narrative analysis.** offers the same map alongside the other narrative tools.

What you get

Output	What it holds
...entity_location_tracking....csv	One row per (character, place, sentence): the pair, the sentence they were named in, and which document it came from. The data behind the map, and the file to open when the map surprises you.
...entity_location_summary....csv	The same information collapsed: how often each character is named with each place.
..._migration_map.html	The animated map. Self-contained: a timeline slider, a character filter, and (from a corpus of several documents) a document filter.

Reading the map

- **The timeline is** the sentence, not a date. The slider runs through narrative order, so "time" here means position in the text.
- **A character's path joins** the sentences in which that character and that place were named together, in order. A line between two points is an inference you are making, not one the tool made.
- **The map shows** only characters and places the recogniser found, and only places Nominatim could place on the globe.

The two things that are capped, and why

Geocoding is the slow step, and Nominatim allows roughly one lookup a second. A corpus with several hundred distinct place names therefore takes several minutes before any map appears.

The Corpus Profiler, which runs unattended, caps the map at the 40 MOST FREQUENT places for that reason. On the Harry Potter corpus that is 40 of 328 distinct places — the map is the busiest places, not all of them. The full tracking CSV is written either way, so nothing is lost from the data; it is the MAP that is abridged.

Run from the GIS GUI there is no cap: every distinct place is looked up, which is what you want when the map matters more than the wait.

What it does not tell you

A character named in the same sentence as a place was not necessarily THERE.

"Harry had never been to Albania" puts Harry and Albania in one sentence. So does "Ron said Egypt was hot". Co-occurrence is a much weaker relation than presence, and on a map it looks exactly like presence. Read the Sentence column of the tracking CSV before

making anything of a surprising point — the sentence is there precisely so the claim can be checked.

Beyond that:

- Only NAMED places are found. "the village", "the castle", "home" are invisible to the recogniser, and they are often where most of the action is.
- Only NAMED characters are found, and each name form is a separate entity here: this tool does not merge "Harry" and "Harry Potter" the way the character emotion arcs do.
- Nominatim geocodes a NAME, not a place in a story: a fictional place with a real-world namesake lands on the real one, and a fictional place with no namesake is silently dropped from the map.
- The animation orders by sentence, and sentence IDs restart in each document. A corpus of many documents is ordered within each document, and the document filter is how you read one narrative at a time.

Getting more out of it

- Use the document filter to follow one text at a time; the whole-corpus view is a summary of places, not a journey.
- Two characters converging on the same place at the same point in the text is the interesting event. The tracking CSV can be filtered for exactly that, and the sentences read.
- For places that cannot be geocoded — rooms, fields, symbolic spaces — use the symbolic-space tool instead. Between them they cover a narrative's geography.

References

Moretti, F. (1998). *Atlas of the European Novel, 1800–1900*. [mapping narrative as a method]

Piatti, B. et al. (2009). "Mapping Literature: Towards a Geography of Fiction." In Cartwright, Gartner & Lehn (eds.), *Cartography and Art*.

Qi, P., Zhang, Y., Zhang, Y., Bolton, J., & Manning, C. D. (2020). "Stanza: A Python Natural Language Processing Toolkit for Many Human Languages." ACL.

See also the NLP Suite TIPS on Extracting locations (NER), Geocoding, Geocoding: How to Improve Nominatim, and Symbolic (non-geocodable) space.